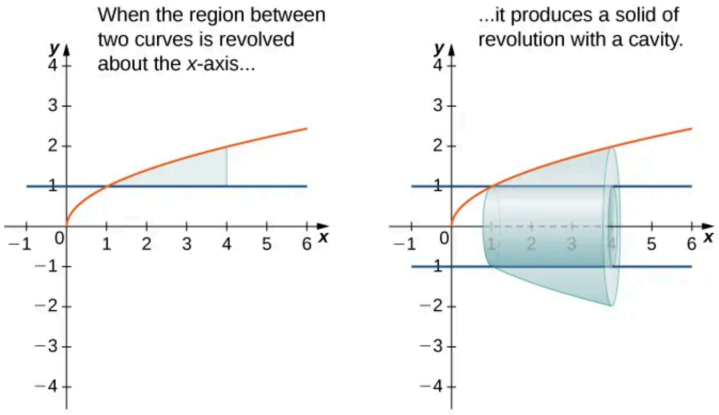
1 The Washer Method

Some solids of revolution have cavities in the middle; they are not solid all the way to the axis of revolution. There are several ways in which this can happen:

- Sometimes, this is just a result of the way the region of revolution is shaped with respect to the axis of revolution.
- In other cases, cavities arise when the region of revolution is defined as the region between the graphs of two functions.
- A third way this can happen is when an axis of revolution other than the x-axis or y-axis is selected.

When the solid of revolution has a cavity in the middle, the slices used to approximate the volume are not disks, but washers (disks with holes in the center).

Example 1.1. Consider the region bounded above by the graph of the function $f(x) = \sqrt{x}$ and below by the graph of the function $g(x) = 1$ over the interval $[1,4]$. When this region is revolved around the x-axis, the result is a solid with a cavity in the middle, and the slices are washers. This is shown in the figure below. The cross-sectional area at $x \in [1,4]$ is then given by the area of the



outer circle minus the area of the inner circle:

$$A(x) = \pi(\sqrt{x})^2 - \pi(1)^2 = \pi(x-1)$$

and the volume of the solid is therefore

$$V = \int_1^4 A(x)dx = \int_1^4 \pi(x-1)dx = \pi\left(\frac{x^2}{2} - x\right)\Big|_1^4 = \frac{9\pi}{2}.$$

Theorem 1.2 (The Washer Method)**.** *Suppose $f(x)$ and $g(x)$ are continuous, nonnegative functions such that $f(x) \geq g(x)$ over $[a,b]$. Let R denote the region bounded above by the graph of $f(x)$, below by the graph of $g(x)$, on the left by the line $x = a$, and on the right by $x = b$. Then, the volume of the solid of revolution formed by revolving R about the x-axis is*

$$\boxed{V = \int_a^b \pi[(f(x))^2 - (g(x))^2]dx.}$$

1. Find the volume of a solid of revolution formed by revolving the region abounded by the graph of $f(x) = x$ and below by the graph of $\frac{1}{x}$ over the interval $[1,4]$ around the x-axis.

2. The region enclosed by the curves $y = x$ and $y = x^2$ is rotated about the x-axis. Find the volume of the resulting solid of revolution.

As with the disk method, we can also apply the washer method to solids of revolution that result from revolving a region about the y-axis.

3. Find the volume of the solid of revolution formed by revolving the region bounded above by $f(x) = 2x^3$ and below by the x-axis over the interval $[0,1]$ about the y-axis.

Let's now consider examples where the axis of revolution is a line other than the x- or y-axis.

4. Find the volume of the solid of revolution formed by revolving the region bounded above by $f(x) = 4 - x$ and below by the x-axis over the interval $[0,4]$ about the line $y = -2$.

5. Find the volume of the solid of revolution obtained by rotation the region in Problem 2 about the line $y = 2$.

6. Find the volume of the solid of revolution obtained by rotation the region from Problem 2 about the line $x = -1$.